Dominic Spencer Hartman

Phoenix, Arizona | DominicSpencerHartman@gmail.com | (530) 559-7393 | LinkedIn Profile | US Citizen

PROFILE: Fourth-Year Electrical Engineering Student at Northern Arizona University with a 3.88 GPA and hands-on experience in systems engineering, predictive modeling, and embedded systems.

EDUCATION

Northern Arizona University (NAU) | Flagstaff, AZ | Bachelor of Science, Electrical Engineering (Senior)

- **Cumulative GPA:** 3.87
- **Honors:** Tau Beta Pi Engineering Honor Society (Fall 2024 – Present)
- **Dean's List:** College of Engineering Dean's List (Fall 2022, Spring 2026)
- **High School:** Bear River H.S, Four years on Varsity Golf Team, Varsity letter

WORK EXPERIENCE & PROJECTS

Northrop Grumman Corporation (NGC) | Chandler, AZ College Intern Technical, Interceptor Systems Team | Summer 2026 – Present

- Raw Binay Stream Processing
- Systems engineering, Block Diagram, Interconnect, and Schematic work
- Electrical Requirements, Compliance Assurance, Mission Assurance
- Cable Design
- Developing Test Procedures

Northrop Grumman Corporation (NGC) | Chandler, AZ College Intern Technical, Medium Launch Vehicles Systems Team | Summer 2025

- Completed a summer 2025 internship supporting the Systems team for Medium Launch Vehicles
- Analyzed rocket telemetry data across multiple Rocket generations
- Developed internal support tool for telemetry analysis

Finalist, ASU Semiconductor Solutions Challenge 2026 | Southwest Advanced Prototyping (SWAP) Hub | – Voltage Drop Prediction Model | April 2026

- Develop and evaluate machine learning models for IR-drop prediction
- Utilize layout-derived features such as current density and power distribution characteristics. Design models with fast inference suitable for integration into design workflows, and provide reproducible implementations, including model artifacts and evaluation scripts
- Semiconductor Engineering, Technical Presentation, Model Evaluation (F1, MAE), Predictive Modeling, Fully Connected Convolutional Network (FCCN), Real Data, PyTorch, ML Semiconductor Problem

American Statistical Association (ASA) DataFest 2023 | Tempe, AZ Data Analyst | March 2023

- Collaborated with three team members to propose an expansion strategy for ABA Free Legal Help

- Utilized Python, pandas, and Tableau to engineer a county-level heat map
- Integrated internal ABA data and US Census poverty data to outline potential areas for expansion to extend coverage to the desired population
- Earned the Judges' Choice Award for our presentation and findings

TECHNICAL SKILLS

- **Software & Programming:** System-Verilog, Cadence, Intel Quartus Prime, Intel Questa, C, Linux, Bash, Python, MATLAB, Git, Arduino, TCL
- **Data & Machine Learning:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib
- **Hardware & Engineering Concepts:** Space Systems, System Design, Digital Signal Processing (DSP), Signals & Systems, Electromagnetics, Cables, Multi-Agent Systems, Photovoltaic Energy, ARM Thumb Instructions, FPGA programming, IoT Design (Embedded Systems), circuit analysis, operational amplifiers, transformers, Diodes, MOSFETs, CMOS, transient analysis, s-domain analysis, Requirements Analysis
- **Certifications:** FCC Ham Radio license, KN6WYS

LEADERSHIP & INVOLVEMENT

- **President:** IEEE Club (Spring 2023 - Fall 2024)
- **Avionics Team Member**: NAU Rocket Club (Fall 2023)